

Chemical Structures in Benzene Derivatives

Benzene

SMILES: c1ccccc1

Formula: C₆H₆ | MW: 78.11 g/mol

Toluene

SMILES: Cc1ccccc1

Formula: C₇H₈ | MW: 92.14 g/mol

Phenol

SMILES: Oc1ccccc1

Formula: C₆H₆O | MW: 94.11 g/mol

Aniline

SMILES: Nc1ccccc1

Formula: C₆H₇N | MW: 93.13 g/mol

Nitrobenzene

SMILES: O=[N+]([O-])c1ccccc1

Formula: C₆H₅NO₂ | MW: 123.11 g/mol

Benzoic acid

SMILES: OC(=O)c1ccccc1

Formula: C₇H₆O₂ | MW: 122.12 g/mol

Acetophenone

SMILES: CC(=O)c1ccccc1

Formula: C₈H₈O | MW: 120.15 g/mol

Styrene

SMILES: C=Cc1ccccc1

Formula: C₈H₈ | MW: 104.15 g/mol

Biphenyl

SMILES: c1ccc(-c2ccccc2)cc1

Formula: C₁₂H₁₀ | MW: 154.21 g/mol

Naphthalene

SMILES: c1ccc2ccccc2c1

Formula: C₁₀H₈ | MW: 128.17 g/mol

Anthracene

SMILES: c1ccc2cc3ccccc3cc2c1

Formula: C₁₄H₁₀ | MW: 178.23 g/mol

Aspirin

SMILES: CC(=O)Oc1ccccc1C(=O)O

Formula: C₉H₈O₄ | MW: 180.16 g/mol

Ibuprofen

SMILES: CC(C)Cc1ccc(C(C)C(=O)O)cc1

Formula: C₁₃H₁₈O₂ | MW: 206.28 g/mol

Caffeine

SMILES: Cn1c(=O)c2c(ncn2C)n(C)c1=O

Formula: C₈H₁₀N₄O₂ | MW: 194.19 g/mol

Cholesterol

SMILES: CC(C)CCCC(C)C1CCC2C1(CCC1C2CC=C2CC(O)CCC21C)C

Formula: C₂₇H₄₆O | MW: 386.65 g/mol

Complex Heterocyclic Structures

Morphine

SMILES: CN1CC[C@]23c4c5ccc(O)c4O[C@H]2C(=C[C@@H]1[C@@H]3O)C=C5

Formula: C₁₇H₁₉NO₃ | MW: 285.34 g/mol

Nicotine

SMILES: CN1CCC[C@H]1c1cccnc1

Formula: C₁₀H₁₄N₂ | MW: 162.23 g/mol

Dopamine

SMILES: NCCc1ccc(O)c(O)c1

Formula: C₈H₁₁NO₂ | MW: 153.18 g/mol

Serotonin

SMILES: NCCc1c[nH]c2ccc(O)cc12

Formula: C₁₀H₁₂N₂O | MW: 176.21 g/mol

Adrenaline (Epinephrine)

SMILES: CNC[C@H](O)c1ccc(O)c(O)c1

Formula: C₉H₁₃NO₃ | MW: 183.20 g/mol

Penicillin G

SMILES: O=C(O)[C@@H]1N2C(=O)[C@@H](NC(=O)Cc3ccccc3)[C@H]2SC1(C)C

Formula: C₁₆H₁₈N₂O₄S | MW: 334.39 g/mol

Glucose

SMILES: OC[C@H]1OC(O)[C@H](O)[C@@H](O)[C@@H]1O

Formula: C₆H₁₂O₆ | MW: 180.16 g/mol

Testosterone

SMILES: C[C@]12CC[C@H]3[C@@H](CCC4=CC(=O)CC[C@@]43C)[C@@H]1CC[C@@H]2O

Formula: C₁₉H₂₈O₂ | MW: 288.42 g/mol

Estradiol

SMILES: C[C@]12CC[C@H]3[C@@H](CCc4cc(O)ccc43)[C@@H]1CC[C@@H]2O

Formula: C₁₈H₂₄O₂ | MW: 272.38 g/mol

Quinoline

SMILES: c1ccc2ncccc2c1

Formula: C₉H₇N | MW: 129.16 g/mol

Isoquinoline

SMILES: c1ccc2cnccc2c1

Formula: C₉H₇N | MW: 129.16 g/mol

Pyridine

SMILES: c1ccncc1

Formula: C₅H₅N | MW: 79.10 g/mol

Indole

SMILES: c1ccc2[nH]ccc2c1

Formula: C₈H₇N | MW: 117.15 g/mol

Purine

SMILES: c1ncc2[nH]cnc2n1

Formula: C₅H₄N₄ | MW: 120.11 g/mol